Leetcode 1781 Sum of Beauty of All Substrings

Monday, March 02, 2026 3:20 AM

Brute Force

```
class Solution:

    def beautySum(self, s: str) -> int:

        n= len(s)

        ans=0

        for i in range(n):

            for j in range(i , n):

                freq= [0] * 26

                for  k in range(i, j+1):

                    freq[ord(s[k]) - ord('a')] +=1
```

Dry Run Brute force

s = "aabcb" , n = len(s)

index 0 1 2 3 4

a a b c d

i = 0

j = 0 → substring "a"

- k loop counts: a = 1
- max = 1 ; min = 1
- beauty = 0

ans = 1

j = 1 - "aa"

- counts a = 2
- max = 2 , min = 2
- beauty = 0

ans = 0

j = 2 → "aab"

- counts a = 2 , b = 1
- max = 2 min = 1
- beauty = 1

i = 1

j = 1 → "a"

j = 3 "aabc"

- counts a = 2, b = 1, c = 1
- max = 2 , min = 1
- beauty = 2
- ans = 3

j = 4 = "aabcb"

- counts a = 2, b = 2, c = 1
- max = 2 , min = 1
- beauty = 1

ans = 3

i = 2

j = 2 , "b"

beauty = 0

i = 3

j = 3 → c

```
mx=0

mn= float("inf")

for f in freq:

    if f>0:

        mx = max(mx,f)

        mn= min(mn,f)

ans+=mx-mn

return ans
```

. a -> 1, beauty = 0
ans = 3
j = 2 -> "ab"
. a = 1, b = 1
ans = 3
j = 3 -> "abc"
. a = 1, b = 1, c = 1
. beauty = 0
ans = 3
j = 4 -> "abcb"
. a = 1, b = 2, c = 1
. max = 2, min = 1
. beauty = 1
ans = 4

ans 4
j = 3 -> bc
. b = 1, c = 1
. beauty = 0
. ans = 4
j = 4 -> bcb
. b = 2, c = 1
. beauty = 1
. ans = 5

. beauty = 0
ans = 5
j = 4 -> "b"
. beauty = 0
. ans = 5
i = 4
j = 4 -> "b"
. beauty = 0
ans = 5

Brute Force

Time Complexity -> $O(n^3)$
Space Complexity -> $O(1)$

Optimal Solution

class Solution:

j = 0

```python
def beautySum(self, s: str) -> int:
    n= len(s)
    result=0

    for i in range(n):
        freq=[0]*26
        for j in range(i,n):
            idx= ord(s[j]) - ord('a')
            freq[idx] +=1
            max_freq=0
            min_freq= float("inf")

            for f in freq:
                if f>0:
                    max_freq= max(max_freq,f)
                    min_freq= min(min_freq,f)
            result += max_freq- min_freq
    return result
```

freq = { }

j = 0 -> "a"

freq -> "a"

beauty = 0

result = 0

j = 1 -> aa

freq = a = 2

beauty = 0

result = 0

j = 2 -> "aab"

freq = a: 2, b=1

beauty = 1

result = 1

j = 3 -> "aabc"

freq = a = 2, b=1, c=1

beauty = 1

result + = 2

j = 4 -> "aabcb"

freq = a = 2, b = 2, c = 1

beauty = 1

result = 3

$i = 1$

Reset freq = {}

$j = 1 \rightarrow$ 'a'

a -> 1, beauty = 0

result = 3

$j = 2 \rightarrow$ "ab"

a = 1, b = 1, beauty = 0

result = 3

, -> "abc"

$i = 2$

$j = 3 \rightarrow$ 'abc'

a = 1, b = 1, c = 1, beauty = 0

result = 3

$j = 4, \rightarrow$ "abcb"

a = 1, b = 2, c = 1, -> beauty = 1

result = 4

Reset freq = { }

j = 2 → "b"

beauty = 0

result = 4

j = 3 → "bc"

beauty = 0

result = 4

j = 4 → bcb

b = 2, c = 1

beauty = 1

result = 5

i = 3

j = 3 → c → 0

Optimal Solution

Time Complexity → $O(n^2)$

Space Complexity → $O(1)$

$j = 4 \rightarrow b \rightarrow 0$

$i = 4$

$j = 4 \rightarrow b \rightarrow 0$

result = 5